

Adem Hmercha

Software Engineer | DevOps

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PROFILE

Software Engineer with an engineering degree in Software Engineering (Esprit), full-stack (Spring Boot/Angular, Node.js/React) and DevOps-oriented (Docker, Kubernetes, CI/CD, GitOps). Final-year internship at Sagemcom: designed a complete platform and deployed an end-to-end DevOps pipeline, from code to production monitoring.

PROFESSIONAL EXPERIENCE

Final-Year Internship — Sagemcom

Jan 2026 – Jul 2026

Telnet Test Manager — web platform for managing and automating Telnet tests on network equipment (routers, switches, embedded gateways)

- Designed and developed a full-stack platform (React 18 + TypeScript / Node.js + Express / MongoDB) to run, monitor, and report Telnet tests in real time across multiple devices simultaneously.
- Built the Telnet execution engine (dedicated Node.js Worker Threads): runs individual commands or full test sequences in parallel across multiple slots/devices without blocking the main server, with live progress tracking via WebSocket.
- Implemented a complete audit-log system (dedicated Audit Logs module in the admin dashboard + /admin/audit-logs endpoint) to track and record all user actions for traceability and compliance.
- Implemented multi-role access control (Admin, Engineer, Technician) with secure JWT authentication, Helmet.js, CORS, and rate limiting.
- Built test reporting with PDF export, full history, and an analytics dashboard; internationalized UI (FR / EN / PT-BR) with react-i18next.
- Designed a full CRUD configuration module (stations, products, slots, references, Telnet commands).
- Set up a complete DevOps pipeline: containerization with Docker and Docker Compose, production orchestration with Kubernetes (Minikube), and configurable deployment via a Helm chart.
- Built a GitHub Actions CI/CD pipeline (backend/frontend build, npm security audit blocking on critical vulnerabilities, Helm deployment with automatic rollback on failure, healthchecks) plus an Nginx reverse proxy.
- Set up application monitoring with Prometheus (custom metrics via prom-client) and real-time Grafana dashboards (backend status, tests launched, active WebSocket connections, latency, CPU/RAM).

Summer Internship — 3S Company

Summer 2024

- Developed a real-estate web application using the MERN stack (MongoDB, Express.js, React, Node.js).
- Implemented property management features, with a responsive front end and a robust backend.

EDUCATION

Engineering Degree in Software Engineering — Esprit

2023 – 2026 (Graduated)

Preparatory Institute for Engineering Studies of Monastir — Mathematics-Physics

2021 – 2023

PROJECTS

RAG Chat — Retrieval-Augmented Generation AI Assistant

Tools: Ollama (Llama 3.2), nomic-embed-text, ChromaDB, MongoDB, Redis, Docker Compose

Code: github.com/ademhmercha/RAG-CHAT

- Fully local and free RAG pipeline: document upload (PDF, DOCX, TXT, CSV, HTML, MD) → chunking → embeddings (Ollama) → vector storage (ChromaDB) → retrieval → answer generation, with background asynchronous ingestion.
- Docker Compose microservices architecture (6 services: Node/Express API, React/Nginx frontend, MongoDB, ChromaDB, Redis, Ollama) with JWT authentication, conversation history, and document management.
- Multi-provider LLM fallback system (local Ollama → Groq → Gemini → OpenRouter) to guarantee service continuity in case of downtime.

User Entity — GitOps Deployment with ArgoCD

Tools: ArgoCD, Kubernetes, Kustomize, Docker, GitHub Actions

Code: github.com/ademhmercha/user-entity-REACT-NODE-JS

- Set up a complete GitOps pipeline with ArgoCD: automatic Kubernetes cluster sync on every push, with self-heal (reverts manual cluster changes) and pruning of resources removed from the Git repo.
- Multi-environment management (dev/prod) via Kustomize (base + overlays) and a GitHub Actions CI/CD pipeline that builds/pushes Docker images and automatically updates the manifests consumed by ArgoCD.
- Secure authentication API (JWT in httpOnly cookie, email verification, password reset) behind an Nginx reverse proxy (SPA fallback + API proxy).

Summary Bot — AI-Powered Telegram Summarization Bot

Tools: Node.js, Google Gemini, pdf-parse, mammoth, cheerio

Code: github.com/ademhmercha/telegram-summarize-bot

- Multi-format summarization (YouTube links, PDF, Word, .txt, web articles, voice messages) via Gemini 2.5, with automatic model fallback and retry logic on quota/rate-limit errors (429/503).
- Advanced features: on-the-fly multilingual translation, follow-up Q&A on the source content, disk-cached results for instant responses, and per-user encrypted personal Gemini API keys (AES-256-GCM).

Discord Moderation Bot

Tools: Node.js, discord.js v14, yt-dlp, ffmpeg

Code: github.com/ademhmercha/discordBot

- Full, traceable moderation system: numbered case IDs for every action, temporary bans with scheduled auto-unban, warnings, and per-server configurable logs.
- Social system (XP, levels, coin economy, auto-unlocked badges) and a music player (persistent queue, automatic resume after restart).

Order Processing — E-Commerce Automation with n8n

Tools: n8n, Express.js, Supabase (PostgreSQL), Docker

Code: github.com/ademhmercha/n8n-order-processing

- 11-node n8n workflow orchestrating the entire order pipeline (validation → stock check → reservation → invoice generation → warehouse task → notification) with conditional success/failure branching.
- Decoupled architecture: a deliberately minimal Express backend (authentication + routing), with all business logic offloaded to n8n and editable without touching application code.

Rakeb — Parental Control Platform (safeguard)

Tools: React, Node.js/Prisma, PostgreSQL (Supabase), React Native/Expo, Firebase

Code: github.com/ademhmercha/safeguard

- Full-stack real-time parental monitoring platform (Supabase Realtime): remote device locking, website/app blocking, browsing history, and instant alerts.
- Child mobile app (React Native/Expo) with 6-character pairing codes, a monitored in-app browser, and screen-time reports; push notifications via Firebase Cloud Messaging.

Kubernetes Auto-Scaling — Node.js, HPA & Metrics Server

Tools: Kubernetes, Docker (multi-stage), Node.js, Metrics Server, Minikube

Code: github.com/ademhmercha/Auto-Scaling-Application

- Express API deployed on Kubernetes with a Horizontal Pod Autoscaler driven by real CPU metrics (Metrics Server), demonstrating auto-scaling from 2 to 10 pods under load with gradual scale-down.
- Multi-stage Docker image (non-root user), liveness/readiness probes, graceful shutdown (SIGTERM, zero dropped requests), and a built-in load-testing script.

GitHub AutoReview Agent — AI-Assisted Code Audit

Tools: React, Vite, Tailwind CSS, Groq API (Llama 3.3 70B)

Code: github.com/ademhmercha/Github-Auto-Review

- Automated public GitHub repository audit tool: 5 specialized AI agents run in sequence (repo exploration, code review, documentation quality, /10 scoring, refactoring suggestions) to produce a complete report.
- 100% frontend architecture (no backend, API calls made directly from the browser), PDF report export, deployed on Vercel.

SKILLS

Frameworks: Angular, Spring Boot, Symfony, .NET, Node.js, React, Express

Languages: Java, Python, C, C#, PHP, JavaScript, TypeScript

DevOps: Docker, Docker Compose, Kubernetes (Helm, HPA, Minikube), ArgoCD, GitOps, Kustomize, Jenkins, GitHub Actions, GitLab CI/CD, Pipelines CI/CD, Prometheus, Grafana, Nginx

Automation & AI: n8n, Gemini API, Groq API, Ollama / local LLMs, RAG

Databases: MongoDB, MySQL, PostgreSQL, Firebase

Management: Agile/Scrum, Jira

LANGUAGES

French: B2 English: B2